

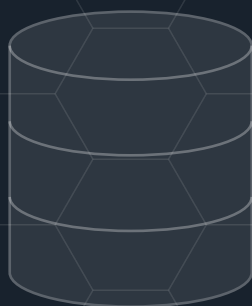
Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 11

Home Subscriber Server (HSS)

سرور مشترک خانگی (HSS)



Lecture Notes

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1 Why a Centralized Subscriber Database?

In any mobile network, thousands to millions of subscribers connect, move, and consume services simultaneously. A centralized subscriber database is essential for:

Function	Purpose
Authentication	Verify subscriber identity before granting network access
Authorization	Determine what services/resources a subscriber is allowed to use
Location Tracking	Know which network element currently serves each subscriber
Subscription Management	Store and enforce service plans, QoS profiles, and restrictions

Without a centralized database:

- Any device could impersonate a subscriber
- The network cannot route incoming calls/data to the correct location
- Service differentiation (premium vs. basic plans) would be impossible
- Roaming between networks would not function

The **HSS (Home Subscriber Server)** is the evolution of the 2G/3G HLR (Home Location Register), re-designed for all-IP LTE/EPC networks using the Diameter protocol instead of SS7/MAP.

2 HSS Role in LTE/EPC

The HSS is the **master database** for all subscriber and subscription data in the Evolved Packet Core (EPC):

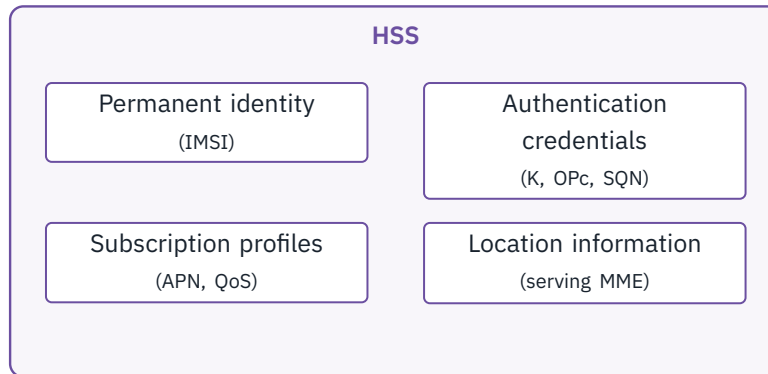


Figure 11.1 What the HSS stores

■ Key Responsibilities:

1. Stores Permanent Subscriber Identity (IMSI)

- International Mobile Subscriber Identity – globally unique 15-digit identifier
- Format: MCC (3 digits) + MNC (2-3 digits) + MSIN (9-10 digits)

2. Stores Authentication Credentials (K, OPc)

- **K**: 128-bit permanent secret key (shared only between USIM and HSS)
- **OPc**: Operator-specific constant derived from OP (operator key) and K
- Never transmitted over the air – used to generate temporary credentials

3. Stores Subscription Profiles (APN, QoS, Allowed Services)

- Which APNs the subscriber can connect to
- Maximum bitrates (AMBR) for uplink/downlink
- Quality of Service class identifiers (QCI)
- Allowed services and restrictions

4. Tracks UE Location (Serving MME)

- Records which MME currently serves the UE
- Enables routing of incoming services (MT-SMS, paging)
- Updated every time UE performs a Tracking Area Update (TAU) or attach

3 HSS Data Model

■ HSS Data Model Diagram

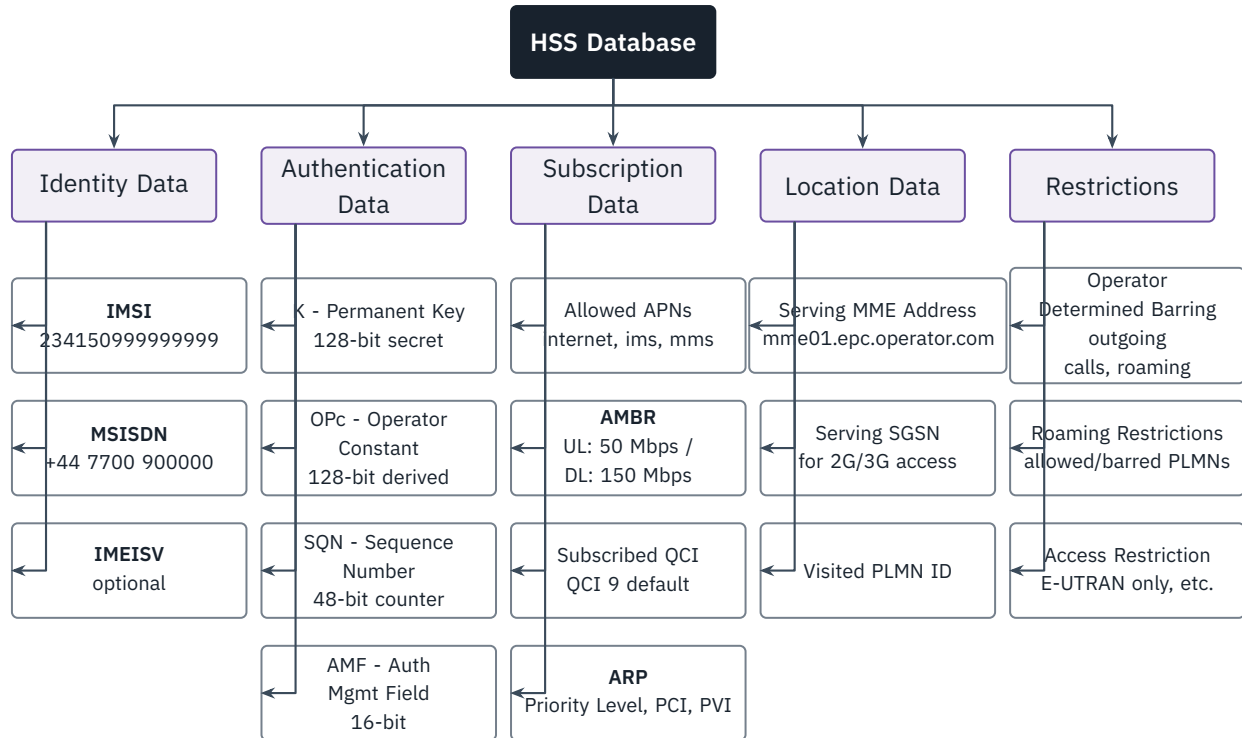


Figure 11.2 HSS data model: identity, authentication, subscription, location and restrictions

Detailed Data Fields

Category	Field	Description	Example
Identity	IMSI	Unique subscriber ID	234150999999999
Identity	MSISDN	Phone number (E.164)	+447700900000
Auth	K	Permanent 128-bit secret key	0x465B5CE8...(hex)
Auth	OPc	Operator variant constant	0xE8ED289D...(hex)
Auth	SQN	Sequence number (anti-replay)	000000000021
Auth	AMF	Authentication Management Field	0x8000
Subscription	Default APN	Primary data connection	"internet"
Subscription	APN list	All allowed APNs	internet, ims, mms
Subscription	UE-AMBR	Max aggregate bitrate	DL:150M, UL:50M
Subscription	APN-AMBR	Per-APN max bitrate	DL:100M, UL:40M
Subscription	QCI	QoS Class Identifier	9 (best effort)
Subscription	ARP	Allocation/Retention Priority	Level 8, PCI=yes, PVI=no
Location	MME Identity	Serving MME hostname/address	mme01.epc.mnc015.mcc234
Location	MME Realm	Diameter realm of MME	epc.mnc015.mcc234.3gppnetwork.org
Location	Purged flag	UE detached/unreachable	true/false
Restrictions	ODB	Operator Determined Barring	Barring of outgoing intl calls
Restrictions	Roaming	Allowed/barred PLMNs	Home only / EU roaming
Restrictions	RAT restriction	Allowed access technologies	E-UTRAN, UTRAN

4 Authentication Vector Generation

■ EPS-AKA (Evolved Packet System - Authentication and Key Agreement)

The HSS generates **Authentication Vectors (AVs)** that enable mutual authentication between UE and network without ever transmitting the permanent key K .

Each Authentication Vector contains:

Component	Size	Purpose
RAND	128 bits	Random challenge (generated fresh by HSS)
AUTN	128 bits	Authentication Token (proves network identity to UE)
XRES	32-128 bits	Expected Response (what UE should reply)
K_{ASME}	256 bits	Base key for deriving all session keys

■ Milenage Algorithm Set

The **Milenage** algorithm set is the standard cryptographic toolkit (3GPP TS 35.205-208):

Inputs: K (128-bit), $RAND$ (128-bit), SQN (48-bit), AMF (16-bit), OPc (128-bit)

Milenage function	Output	Purpose
$f1(K, RAND, SQN, AMF)$	MAC-A	Network authentication
$f1^*(K, RAND, SQN, AMF)$	MAC-S	Resynchronisation
$f2(K, RAND)$	RES / XRES	Response
$f3(K, RAND)$	CK	Cipher key
$f4(K, RAND)$	IK	Integrity key
$f5(K, RAND)$	AK	Anonymity key
$f5^*(K, RAND)$	AK	Anonymity key for resynchronisation

Vector derivation process:

1. HSS generates random 128-bit **RAND**
2. HSS increments **SQN** (sequence number)
3. Compute: $MAC-A = f1(K, RAND, SQN, AMF)$
4. Compute: $XRES = f2(K, RAND)$
5. Compute: $CK = f3(K, RAND)$
6. Compute: $IK = f4(K, RAND)$
7. Compute: $AK = f5(K, RAND)$
8. Construct: $AUTN = (SQN \oplus AK) || AMF || MAC-A$
9. Derive: $KASME = KDF(CK, IK, SN-ID, SQN \oplus AK)$

■ SQN Management (Replay Protection)

- **SQN** is a 48-bit counter maintained independently by HSS and USIM
- Prevents replay of old authentication vectors
- UE verifies: received SQN > last accepted SQN (within a window)
- If SQN is out of range → UE sends **AUTS** (resynchronization token)
- HSS uses f1* and f5* to resynchronize its SQN with the USIM

■ Vector Delivery to MME

- HSS typically generates **multiple vectors** (batch) per request
- MME stores unused vectors for subsequent re-authentications
- Reduces signaling load (MME doesn't need to contact HSS every time)
- Typical batch size: 3-5 vectors per AIR request

5 HSS Interaction with MME (S6a / Diameter)

The **S6a interface** uses the **Diameter protocol** (RFC 6733) with the 3GPP Diameter application for EPS (application ID: 16777251).

■ Message Flows

Procedure	Request → Answer	Direction	Purpose
Authentication	AIR → AIA	MME → HSS	Get auth vectors
Location Update	ULR → ULA	MME → HSS	Register MME, get subscription
Purge	PUR → PUA	MME → HSS	UE explicitly detached
Cancel Location	CLR → CLA	HSS → MME	Force detach (e.g., UE moved)
Insert Sub Data	IDR → IDA	HSS → MME	Push subscription changes
Delete Sub Data	DSR → DSA	HSS → MME	Remove subscription data
Notification	NOR → NOA	MME → HSS	Notify terminal info

■ Attach Procedure (AIR/AIA + ULR/ULA)

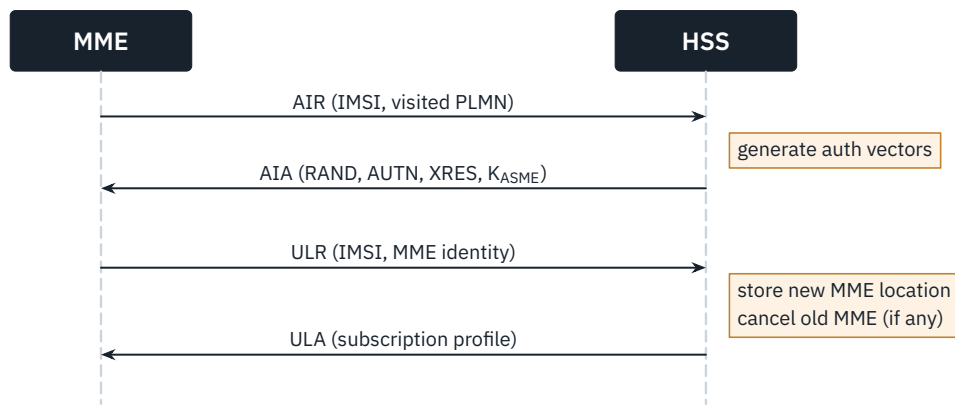


Figure 11.3 S6a exchange between the MME and the HSS during attach

AIR (Authentication-Information-Request):

- IMSI of the subscriber
- Visited PLMN ID
- Number of requested vectors
- Resynchronization info (AUTS, if SQN mismatch)

AIA (Authentication-Information-Answer):

- One or more EPS Authentication Vectors (RAND, AUTN, XRES, K_{ASME})
- Result code (success/failure)

ULR (Update-Location-Request):

- IMSI
- MME identity (hostname + realm)
- RAT type
- ULR flags (initial attach, etc.)

ULA (Update-Location-Answer):

- Subscription data (all APNs, AMBR, QoS profiles)
- Result code
- Separation of default APN from additional APNs

■ Detach / Purge (PUR/PUA or CLR/CLA)

Explicit detach (UE-initiated):

- MME sends **PUR** → HSS marks subscriber as “not attached”
- HSS responds with **PUA**

Network-initiated cancel (HSS pushes):

- HSS sends **CLR** to old MME (e.g., UE registered at new MME)
- Old MME removes UE context, responds with **CLA**
- CLR reasons: UE moved, subscription withdrawn, operator policy

■ Subscription Modification (IDR/IDA)

When operator modifies a subscription (e.g., customer upgrades plan):

- HSS sends **IDR** (Insert-Subscriber-Data-Request) to serving MME
- MME updates local copy of subscription data
- MME responds with **IDA**
- Changes take effect immediately (no re-attach needed)

6 HSS Interaction with Other Network Elements

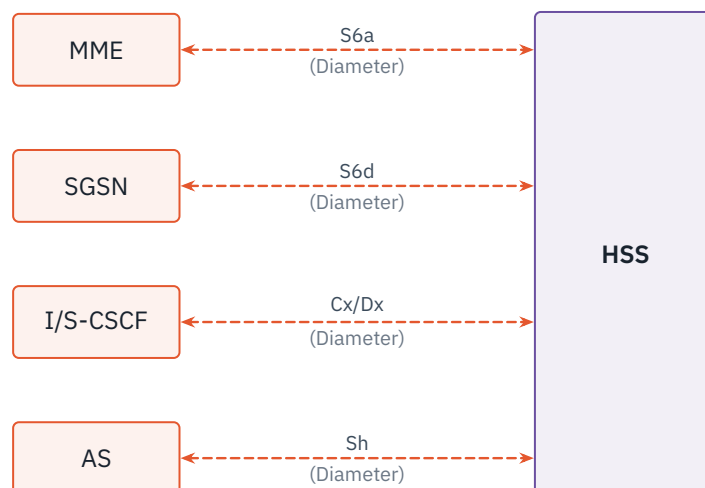


Figure 11.4 Diameter interfaces of the HSS

Interface Details

Interface	Peer Entity	Protocol	Purpose
S6a	MME	Diameter	EPS authentication, location, subscription
S6d	SGSN	Diameter	2G/3G interworking (same functions as S6a for GERAN/UTRAN)
Cx	I-CSCF / S-CSCF	Diameter	IMS registration, user authorization
Dx	I-CSCF	Diameter	IMS location query (find S-CSCF for user)
Sh	Application Server	Diameter	Read/write user data for services
S13	EIR	Diameter	Equipment identity check (IMEI verification)

S6d (SGSN – 2G/3G interworking)

- Same message set as S6a but adapted for GERAN/UTRAN access
- Used when UE is connected via 2G/3G radio
- Enables seamless handover between 4G and 2G/3G (IRAT)

Cx/Dx (IMS – VoLTE)

- Cx:** Registration/de-registration of IMS users
 - UAR/UAA: User-Authorization (during REGISTER)
 - SAR/SAA: Server-Assignment (assign S-CSCF)
 - MAR/MAA: Multimedia-Auth (get IMS auth vectors)
 - LIR/LIA: Location-Info (find assigned S-CSCF)
- Used for VoLTE call setup and IMS registration

Sh (Application Servers)

- Allows AS to read/update user-specific data
- Used for supplementary services, presence, messaging
- Supports subscriptions for notifications on data changes

7 Authentication Scenario (Step-by-Step)

Full EPS-AKA Authentication Flow

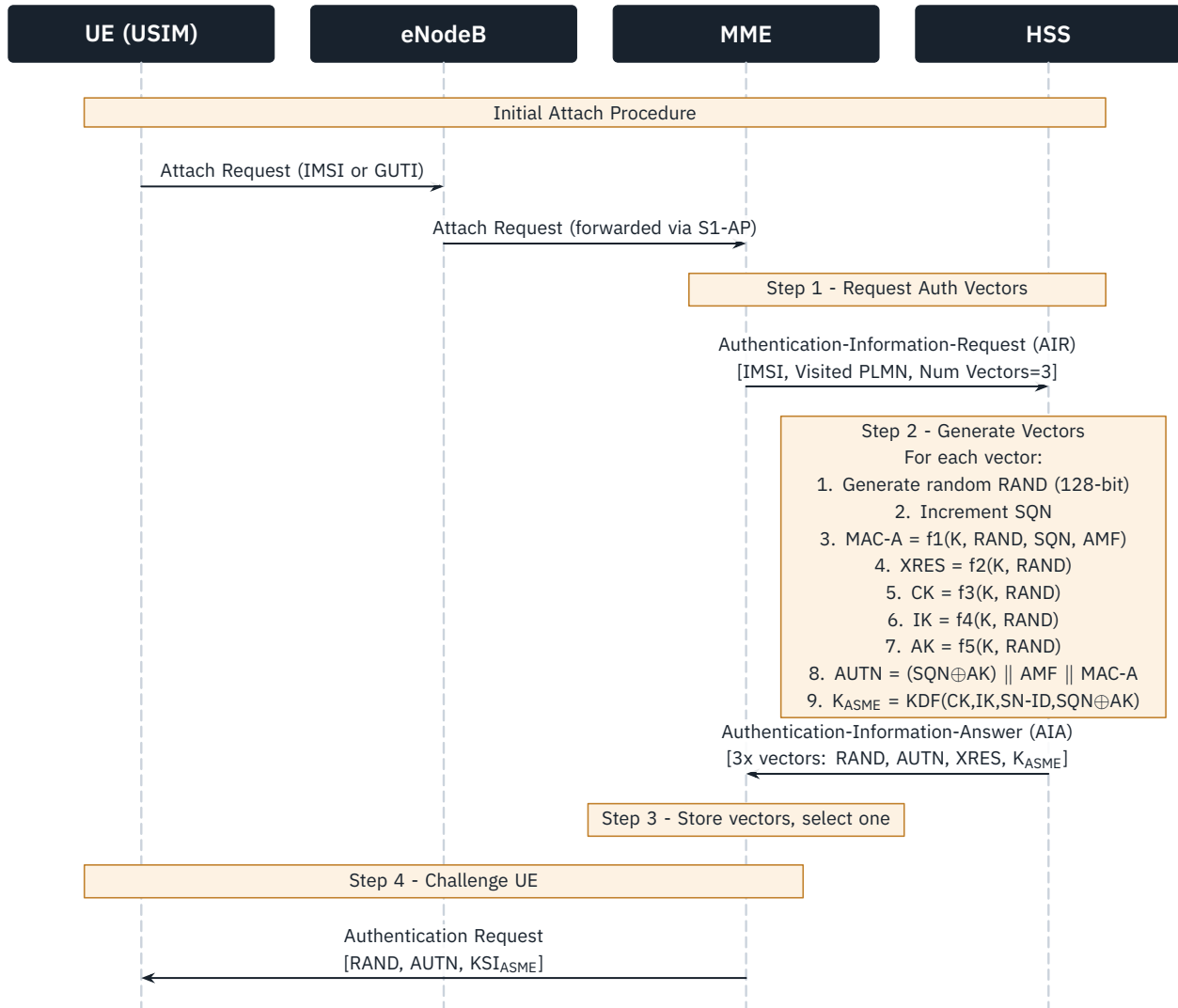


Figure 11.5 Full EPS-AKA authentication flow with the key computations (part 1 of 2)

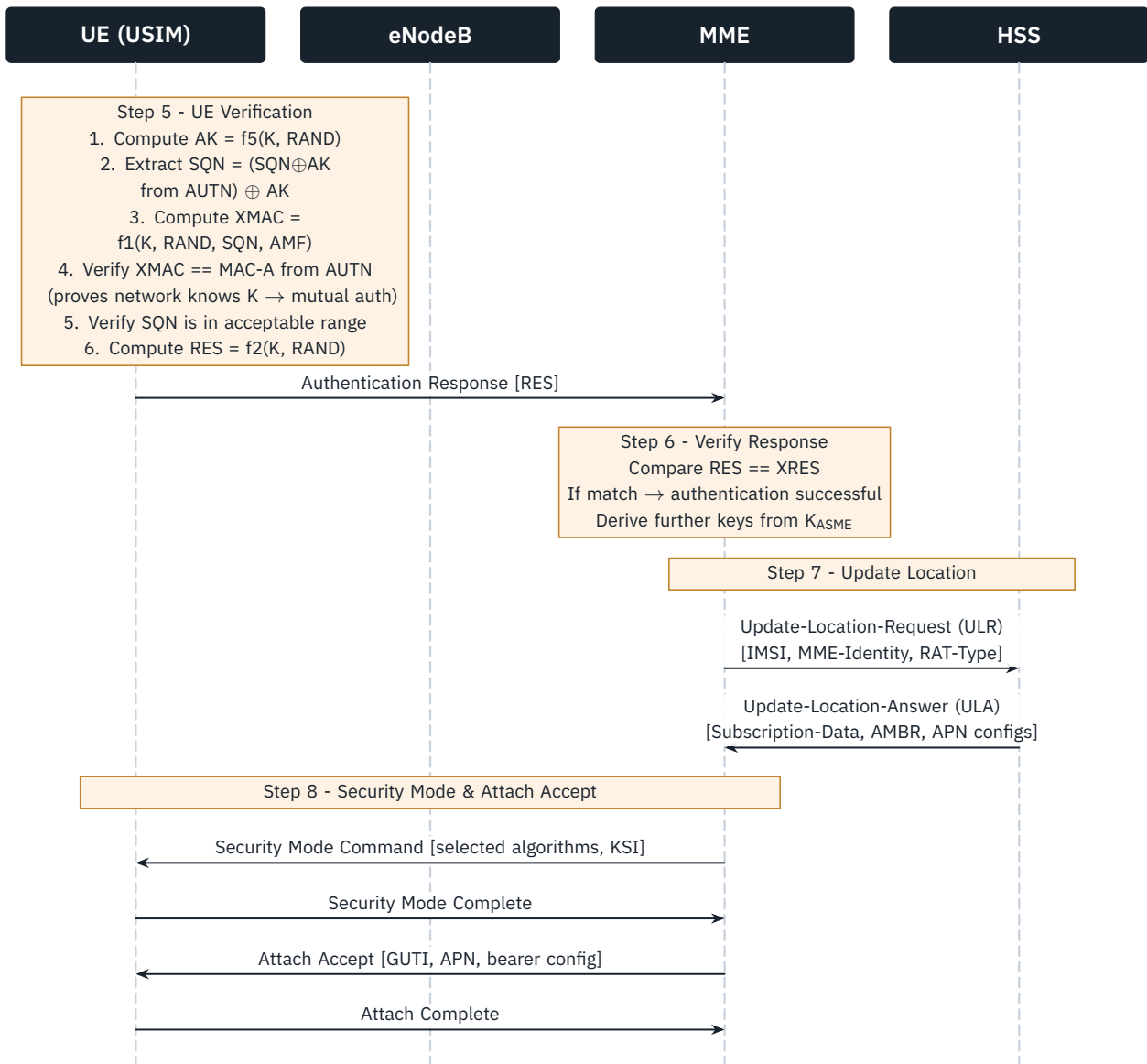


Figure 11.6 Full EPS-AKA authentication flow with the key computations (part 2 of 2)

Step-by-Step Explanation

Step	Action	Details
1	UE sends Attach Request	Contains IMSI (first time) or old GUTI (subsequent)
2	MME requests vectors	AIR sent to HSS with subscriber IMSI
3	HSS generates vectors	Uses $K + OPc + RAND + SQN$ through Milenage
4	HSS returns vectors	AIA contains $RAND, AUTN, XRES, K_{ASME}$ (batch of 3-5)
5	MME challenges UE	Sends $RAND + AUTN$ to UE in NAS Authentication Request
6	UE verifies network	Checks MAC in AUTN using its own $K \rightarrow$ mutual authentication
7	UE computes RES	Sends response back to MME
8	MME verifies UE	Compares RES against XRES from HSS
9	Location update	MME registers itself at HSS as serving MME
10	Session setup	Keys derived, bearers established

Authentication Failure Scenarios

Scenario	Cause	Recovery
MAC failure at UE	Network doesn't know correct K	UE sends Auth Failure (cause: MAC failure)
SQN out of range	SQN desynchronized	UE sends Auth Failure + AUTS for resync
XRES mismatch at MME	UE doesn't know correct K	MME rejects attach, may send Auth Reject
Timeout	Network/UE unreachable	Retry with backoff

8 Evolution to 5G: HSS Decomposition

Architecture Comparison

In 5G (3GPP Release 15+), the monolithic HSS is decomposed into specialized Network Functions:

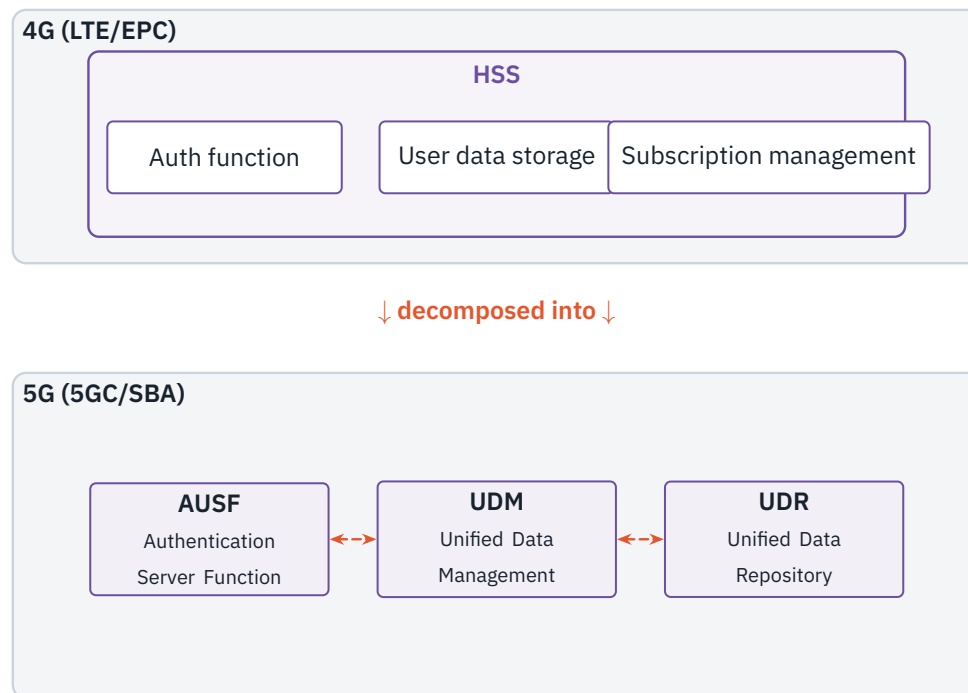


Figure 11.7 The HSS is decomposed into AUSF, UDM and UDR in 5G

Detailed Comparison: HSS (4G) vs UDM + AUSF + UDR (5G)

Aspect	HSS (4G)	UDM + AUSF + UDR (5G)
Architecture	Monolithic	Decomposed, microservices
Protocol	Diameter (S6a)	HTTP/2 + JSON (SBI)
Auth Function	Built into HSS	Separate AUSF (Authentication Server Function)
Data Management	Built into HSS	Separate UDM (Unified Data Management)
Data Storage	Internal DB	Separate UDR (Unified Data Repository)
Identity	IMSI (sent in clear)	SUPI/SUCI (SUCI = encrypted SUPI)
Privacy	IMSI exposed on air	SUCI conceals identity (ECIES encryption)
Auth Protocol	EPS-AKA only	5G-AKA or EAP-AKA' (extensible)
Key Hierarchy	$K_{ASME} \rightarrow K_{eNB} \rightarrow \dots$	$K_{AUSF} \rightarrow K_{SEAF} \rightarrow K_{AMF} \rightarrow K_{gNB} \rightarrow \dots$
Scalability	Scale entire HSS	Scale each NF independently
Interfaces	Point-to-point Diameter	Service-Based Interface (REST APIs)
Discovery	Static config / DNS	NRF (Network Repository Function)
Redundancy	Active/Standby	Cloud-native, stateless NFs + shared UDR

5G Identity: SUPI and SUCI

SUPI (Subscription Permanent Identifier)
 = IMSI (same format as 4G: MCC+MNC+MSIN)
 (never sent over the air in 5G!)

SUCI (Subscription Concealed Identifier)
 = Home Network ID + Routing Indicator + Protection Scheme + Encrypted MSIN

Encryption: ECIES (Elliptic Curve Integrated Encryption Scheme)

- Home network publishes public key
- UE encrypts MSIN portion with home network public key
- Only home network (UDM) can decrypt → reveals SUPI

5G Authentication Methods

5G-AKA:

- Enhanced version of EPS-AKA
- Adds home network confirmation (HRES/HXRES)
- Serving network proves to home network that auth completed
- Prevents certain MITM attacks possible in 4G

EAP-AKA':

- Extensible Authentication Protocol variant

- Binds authentication to serving network name
- Better suited for non-3GPP access (Wi-Fi, fixed)
- Uses AT_KDF_INPUT for network binding

5G Decomposed Functions

5G NF	Role	4G Equivalent
AUSF	Handles authentication procedures, generates keys	HSS auth function
UDM	Manages subscription data, generates auth vectors, handles registration	HSS data management
UDR	Stores all data (subscriber, policy, application)	HSS internal database
SIDF	Decrypts SUCI → SUPI (part of UDM)	N/A (no equivalent in 4G)

Summary

The HSS is the security and identity anchor of the LTE/EPC network. It:

1. **Guards the keys** — K never leaves the HSS (and USIM)
2. **Proves identity** — Mutual authentication via EPS-AKA
3. **Controls access** — Subscription profiles determine what users can do
4. **Tracks location** — Knows which MME serves each subscriber
5. **Evolves gracefully** — Decomposed into UDM/AUSF/UDR in 5G for cloud-native deployment

Understanding the HSS is critical for:

- Network security analysis
- Debugging authentication failures
- Planning subscriber provisioning
- Understanding the transition from 4G to 5G core

References

- 3GPP TS 29.272: S6a/S6d Diameter-based interface (MME/SGSN–HSS)
- 3GPP TS 33.401: EPS security architecture
- 3GPP TS 35.205-208: Milenage algorithm specification
- 3GPP TS 23.501: 5G System Architecture
- 3GPP TS 33.501: 5G Security Architecture and Procedures
- 3GPP TS 29.509: AUSF Services (5G)
- 3GPP TS 29.503: UDM Services (5G)